

operate in poor weather and to carry substantial loads. However, while airplanes struggled to get off the ground with more than two people on board, airships measured their payloads in tons.

Aerial monsters

As far as most Germans were concerned, the dominant form of aircraft before 1914 was the Zeppelin, named after its creator, Count Ferdinand von Zeppelin. Although airships were also developed in other countries, it was only in Germany that they attained the status of a national icon.

Zeppelins were “rigid” airships – that is, the shape of the hydrogen-filled envelope was maintained by a solid framework rather than by the pressure of the gas inside. The LZ 1, von Zeppelin’s first airship, tested in 1900, was 420ft (128m) long. Eventually these aerial monsters would grow to almost twice that size. Inevitably, such dimensions made them hard to handle on the ground, and expensive to manufacture – it was estimated that in 1914 you could make 34 airplanes for the cost of one Zeppelin.

Germany’s love affair with the Zeppelin really began in 1906 when the count first achieved sustained flight with his LZ 3. Carrying out flights of up to eight hours’ duration, he won official support from both the army and the royal family. Patriotic fervor rose to fever pitch when Zeppelin’s next airship, LZ 4, was destroyed on the ground by a storm during a highly publicized journey up the Rhine valley in 1908. The German public spontaneously underwrote the sum of six million marks to allow von Zeppelin to continue his work, and Zeppelin airships began passenger services

AIRSHIP PASSENGERS
The LZ 11 “Viktoria Luise” produced at Zeppelin’s Friedrichshafen base, made its first passenger flight on March 4, 1912. In total it made 1,000 trips, flying between Hamburg, Heligoland, and Copenhagen.

in 1910. Under the direction of chief engineer Ludwig Durr, Zeppelin design steadily improved. From 1912, goldbeater’s skin – a fine membrane from a cow’s intestine – replaced rubberized cotton as the material used to make the gas cells inside the airship’s envelope. It was lighter and removed the risk of igniting the hydrogen with static electricity that could be generated by cotton surfaces rubbing together. For the framework, a new aluminum alloy called duralumin was in use by 1914, offering the strength of steel at one third of the weight. These improvements, plus more powerful engines, allowed the LZ 26 to carry a 14.2-ton (13-tonne) load at more than 50mph (80kph).

Instruments of war

Although Zeppelins had carried more than 37,000 passengers by 1914, passenger transportation in airplanes was an idea whose day had yet to come. They were to find their first practical use in war. In the summer of 1911, a standoff between France and Germany over their interests in Morocco brought Europe to the brink of war. The crisis was resolved after much saber-rattling, but a major conflict seemed likely – a question not of whether, but of when. Stirred up by a jingoistic press, popular opinion demanded that

FLYING IN TO HAMBURG

The LZ 13 “Hansa” is pictured looming over Hamburg harbor in 1912 on the last leg of a journey from Scandinavia. From 1910, Zeppelins made regular flights and carried over 37,000 passengers before World War I, mostly on sightseeing trips.

COUNT FERDINAND VON ZEPPELIN

AN UNLIKELY AERIAL PIONEER, Ferdinand von Zeppelin (1838–1917) was born into the military aristocracy, and served as a cavalry officer until his fifties. His interest in airships was inspired by a visit to the United States during the Civil War, where he witnessed the use of tethered balloons as military observation posts. From 1891, he devoted his personal fortune to the development of powered rigid airships. Despite numerous setbacks, his first airship, LZ 1, made its maiden voyage on July 2,



NATIONAL HERO
The success of Count von Zeppelin’s airships revived national pride and made him a celebrity.

1900. When the LZ 4 was destroyed by a storm in 1908, the popular response revived von Zeppelin’s fortunes. He lived long enough to see his airships being used as bombers during World War I.

AIRSHIP DISPLAY

This German poster of 1913 depicts a heaving crowd attending a Zeppelin airshow. From 1908, Zeppelins made routine flights, carrying mail and passengers throughout Germany.

